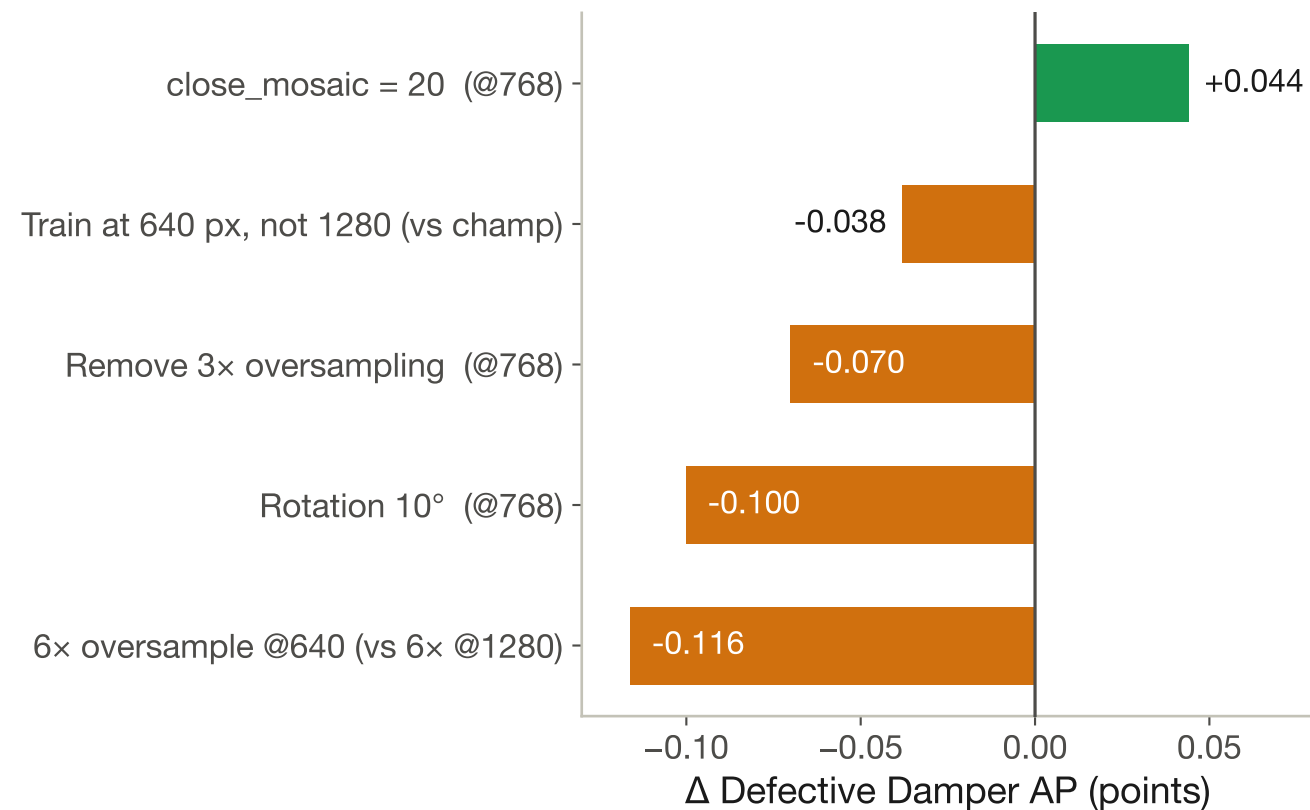
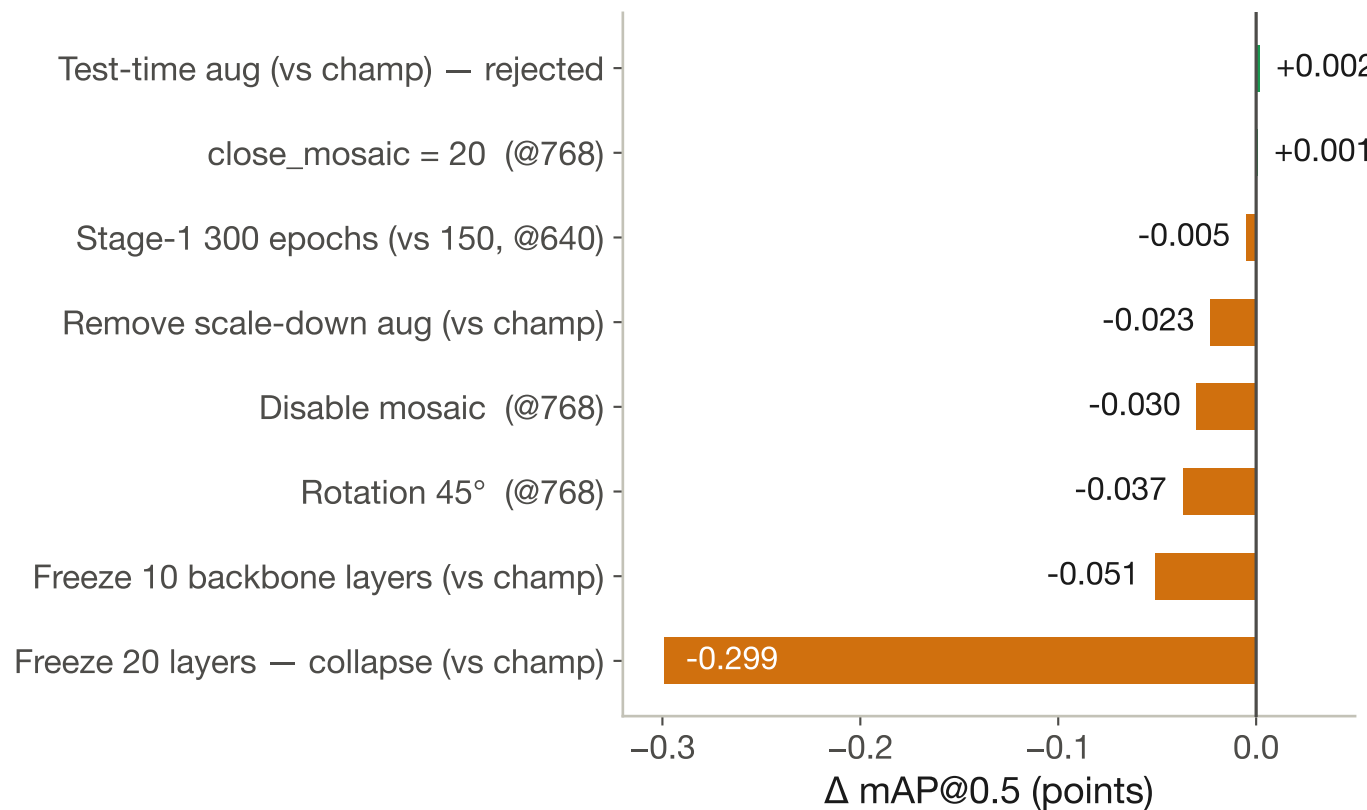


What moves the rare class: resolution + sampling help; freezing and rotation hurt

Ablations of the recipe · each bar states its reference: 'vs champ' = fixed split @1280 vs the full recipe · '@768' = clean benchmark vs the deployment recipe, 3 seeds/arm



Copy-paste augmentation also hurt the rare class (paste artifacts; no seed-averaged delta logged). Scale-jitter sweet spot: 0.85–0.9.